they can happen in any one of the infinite points of a spatial continuum. This is why the result can be infinite.

When quantum gravity is taken into account, these infinities also disappear. The reason is clear: space is not infinitely divisible, there are no infinite points; there are no infinite things to add up. The granular discrete structure of space resolves the difficulties of the quantum theory of fields, eliminating the infinities by which it is afflicted.

This is a tremendous result: on the one hand, taking quantum mechanics into account resolves the problems generated by the infinities of Einstein's theory of gravity, that is to say, the singularities. On the other, taking gravity into account solves the problems generated by quantum field theory, that is to say, the divergences. Far from being contradictory, as they at first seemed, the two theories each offer the solution to the problems posed by the other!

Putting a limit to infinity is a recurrent theme in modern physics. Special relativity may be summarized as the discovery that there exists a maximum velocity for all physical systems. Quantum mechanics can be summarized as the discovery that there exists a maximum of information for each physical system. The minimum length is the Planck length *LP*, the maximum velocity is the speed of light *c*, and the total information is determined by the Planck constant *h*. This is summarized in table 11.1.

The existence of these minimum and maximum values for length, velocity and action fixes a natural system of units. Instead of measuring speed in kilometres per hour, or in metres per second, we can measure it in fractions of the speed of light. We can fix the value 1 for the velocity *c* and write, for example, $v = ½$, for a body which is moving at half the speed of light. In the same way, we can posit by definition and measure length in multiples of Planck's length. And we can posit $h = 1$ and measure actions in multiples of Planck's constant. In this way, we have a natural system of fundamental unities from which the others follow. The unity of time is the time that light takes to cover the Planck length, and so on. The natural unities are commonly used in research on quantum gravity.

The identification of these three fundamental constants places a limit to what seemed to be the infinite possibilities of nature. It suggests that what we call infinite often is nothing more than something which we have not yet counted, or understood. I think this is true in general. 'Infinite', ultimately, is the name that we give to what we do not yet know. Nature appears to be telling us that there is nothing truly infinite.

Table 11.1 Fundamental limitations discovered by theoretical physics.

Physical Quantity	Fundamental constant	Theory	Discovery